

2020 AMC 10A Problem 24 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 24. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 24

Problem:

Let n be the least positive integer greater than 1000 for which

$$\gcd(63, n + 120) = 21 \quad \text{and} \quad \gcd(n + 63, 120) = 60$$

What is the sum of the digits of n ?

Answer Choices:

- A. 12
- B. 15
- C. 18
- D. 21
- E. 24

Solution:

We know that

$$\gcd(n + 57, 63) = 21$$

and

$$\gcd(n - 57, 120) = 60$$

by the Euclidean Algorithm. Hence, let

$$n + 57 = 21\alpha \text{ and } n - 57 = 60\gamma, \text{ where } \gcd(\alpha, 3) = 1 \text{ and } \gcd(\gamma, 2) = 1$$

Subtracting the two equations, $38 = 7\alpha - 20\gamma$. Letting $\gamma = 2s + 1$, we get $58 = 7\alpha - 40$. Taking modulo 40, we have $\alpha \equiv 14 \pmod{40}$. We are given that $n = 21\alpha - 57 > 1000$, so $\alpha \geq 51$. Notice that if $\alpha = 54$ then the condition $\gcd(\alpha, 3) = 1$ is violated. The next possible value of $\alpha = 94$ satisfies the given condition, giving us $n = 1917$.

Alternatively, we could have said $\alpha = 40k + 14 \equiv 0 \pmod{3}$ for $k \equiv 1 \pmod{3}$ only, so $k \equiv 0, 2 \pmod{3}$, giving us our answer. Since the problem asks for the sum of the digits of n , our answer is $1 + 9 + 1 + 7 = \boxed{(C) 18}$.

OR

The conditions of the problem reduce to the following: $n + 120 = 21k$ and $n + 63 = 60\ell$, where $\gcd(k, 3) = 1$ and $\gcd(\ell, 2) = 1$. From these equations, we see that $21k - 60\ell = 57$. Solving this Diophantine equation gives us that $k = 20a + 17$ and $\ell = 7a + 5$. Since, $n > 1000$, we can do some bounding and get that $k > 53$ and $\ell > 17$. Now we start bashing by plugging in numbers that satisfy these conditions: $a = 4$ is the first number that works so we get $\ell = 33, k = 97$. Therefore, we have $n = 21(97) - 120 = 60(33) - 63 = 1917$, and our answer is $1 + 9 + 1 + 7 = \boxed{(C) 18}$.

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